

A Novel Projection Screen using the Crystalline Film of a Frozen Soap Bubble

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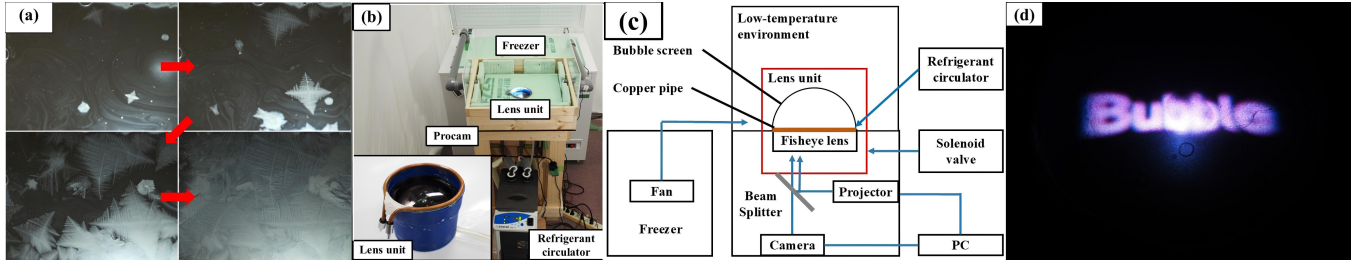


Figure 1: (a) The beautiful and unique freezing process of the soap film. (b) Prototype setup for freezing soap film. (c) System diagram of soap film freezing screen. (d) Freezing and projection by actual equipment."

Abstract

This study proposes a novel screen that focuses on the freezing phenomenon of soap films. Although soap films are thin and transparent, in low-temperature environments, beautiful ice crystals form on the surface, reducing transparency and enabling image projection onto the film. This study explores the application of soap films as multi-layered projection screens that leverage the aesthetic qualities of the freezing process, and discusses potential expansion technologies. We also explore the possibility of applying frozen soap films as a novel screen.

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1 INTRODUCTION

Soap bubbles have long fascinated people with their interference patterns, changes in shape, fragility, and feelings of floating. Their unique beauty has long been a source of fascination. In the fields of science and engineering, many studies have been conducted to explain this beautiful phenomenon [Ahmadi et al. 2019] and to explore ways to apply it [Korenjak and Humar 2023].

In the field of Human-Computer Interaction (HCI), efforts have been made to use the unique beauty of soap films as projection

screens by employing various technical methods. Due to the transparency of soap films, it is challenging to project images onto them as light passes through. However, techniques such as creating surface irregularities using ultrasonic waves [Ochiai et al. 2013] or injecting smoke into the film [Seah et al. 2014] have made it possible to project images onto soap films or bubbles, creating beautiful screens. In this study, we propose a new approach to further explore the potential of soap film screens. Our approach focuses on the reduced transparency of soap films when they are frozen, transforming them into projection screens.

The first notable feature of the frozen soap film screens is the unique crystalline patterns that appear after freezing. These patterns make the transparency of the frozen soap film screen uneven, which makes it unsuitable for applications that require uniform projection, like conventional projector screens. However, the variety of patterns formed during the freezing process adds a highly aesthetic visual element. Thus, this study focuses on the potential for creating beautiful visual displays by combining these crystalline patterns with projected images.

The second feature is the unique and visually striking freezing phenomenon that appears during the membrane's freezing process, along with its potential application as a multi-layered screen. When a soap film is exposed to temperatures below the freezing point of the soap solution, the freezing process begins with the appearance of ice fragments on the membrane. These ice fragments form when airborne dust or surface frost attaches to the supercooled film, acting as a nucleus for freezing. Depending on various factors, such as environmental temperature, the location, shape, growth rate, and movement of the ice fragments differ significantly each time, resulting in varied visual effects. Especially, in environments where there is a temperature difference across the soap film, the freezing process causes convection in the liquid, which makes the ice fragments move across the film. This creates a visually stunning phenomenon

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called the "snow globe effect"[Ahmadi et al. 2019](Figure 1(a)). The soap film becomes fully frozen when these ice fragments increase in number and size until they eventually cover the entire surface. We focus on the unique properties of the soap film during the freezing process, where both highly transparent areas (unfrozen regions) and less transparent areas (frozen regions or ice fragments) coexist on the film. This gives the frozen soap film a distinctive quality as a projection screen. Additionally, we explore how these visual characteristics dynamically change in response to natural phenomena, such as temperature conditions.

In this study, we reproduced the freezing phenomenon of soap films and constructed a system that enables image projection. This paper describes the initial prototype developed and discusses potential applications and possibilities for future expansion.

2 PROTOTYPING

The appearance of the prototype produced is shown in Figure 1(b), and the system configuration is shown in Figure 1(c). The prototype consists of two parts: a soap film generation/cooling system and an all-surrounding projection system.

First, the soap film generation mechanism consists of a 110 mm diameter copper pipe covered with soap film and a small pump that supplies air to the inside of the film, causing it to expand from a flat surface into a dome shape approximately 50 mm high. At the same time, to cool the soap film from the outside, a fan placed around it blows cooled air onto the soap film. In this prototype, the temperature at the bottom of the soap film can be independently adjusted in order to create the temperature difference between the bottom and the top of the film necessary to generate convection currents for the snow dome effect.

To ensure safety, this prototype did not use liquefied gases such as carbon dioxide or nitrogen for cooling. Instead, a freezer capable of cooling down to -60°C was employed. This cooling mechanism allowed the temperature in the space to be lowered to -10°C . The cooling water in the copper pipe was stabilized at -2°C using a cooling water circulation system.

Next, images were projected onto the soap film by an omnidirectional projection system placed under the soap film to allow projection from within the soap film onto the entire surface. This projection system consisted of a projector (ViewSonic PX748-4K), camera (FLIR BFS-U3-13Y3M), beam splitter and fish-eye lens (Opteka Vortex). This setup allows for simultaneous 180° omnidirectional projection in the soap film and omnidirectional imaging with a coaxial camera. Next, Figure 1(d) shows the actual freezing of the soap film and video projection using the prototype we produced. In this prototype, freezing was confirmed at a refrigerant temperature of -2°C at the base and -10°C in the surrounding area. The soap solution used was a common one, prepared with 45% water by mass, 10% commercial dishwashing detergent, and 45% polyvinyl alcohol. The soap film was formed by rubbing a string soaked in the soap solution against a copper pipe by hand.

3 APPLICATION AND DISCUSSION

This section outlines potential applications of the proposed system. The proposed system offers a way to visually enhance the aesthetic qualities of the soap film's freezing phenomenon. This can be achieved by detecting the position and shape of ice fragments in

the soap film and generating projected effect images according to their shape and movement. In this case, we believe it is also possible to create an immersive space by multilayered projection of the soap film and the surrounding environment, since images projected onto the transparent, unfrozen areas will pass through the film. In addition, it is possible to extract ice flake areas on the membrane through image processing of the camera image, dynamically generate a mask image, and combine it with the projected image in real time. As a result, it is possible to achieve high-contrast projection by projecting images only in the ice flake area, or to avoid the ice flakes on the membrane and project images only in the surrounding environment.

Next, since the soap film is a thin film, it can be thawed and become transparent by giving it low thermal energy. We believe that this property can be used to enable interaction with the frozen soap film by the system and the user. For example, the screen can be partially thawed by bringing the user's finger close to it, by direct contact, by blowing air on it, or by shining a laser pointer on it. Furthermore, if thawing could be done using projector light, the system could potentially create multilayered screens with precise and selective transparency in any shape.

4 FUTURE WORK

In this study, we proposed a new projection technology that enables direct image projection onto soap films by freezing them. Furthermore, we developed a prototype that can reproduce the "snow globe" phenomenon.

In the future, we will evaluate the performance of frozen soap film as a screen compared to conventional screen materials. We will also conduct freezing experiments under various conditions to investigate how different temperature conditions affect the patterns on the film. Furthermore, we plan to summarize the visual changes when images are projected onto the frozen films.

Furthermore, efforts will be made to improve the prototype for more stable freezing and to develop practical applications for the system. At the same time, we aim to further evolve the technology so that the freezing process can be controlled by the system and the user, by dynamically controlling convection on the film through partial temperature control and by adding external factors that affect freezing, such as dust in the air. We also plan to investigate projection systems that incorporate frozen soap films in various shapes beyond flat or spherical surfaces, as well as dynamic shape-deformation elements in the soap films.

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